

Divergence-form and wide-halo mEVP in the GPU version of FESOM2: implementation and measurements

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6 August 2026

1. Scope

Two of your proposals were implemented in our GPU version of FESOM2 and measured: **(a)** the **reformulation of the mEVP sub-cycling** in which the quantity carried from one sub-cycle to the next is the divergence of the stress, held at nodes, instead of the three stress components held at elements; and **(b)** the **wide halo**, in which the halo is exchanged only every K -th sub-cycle and the missing information is recomputed locally in a ring of ghost cells K deep, so that the result is identical to exchanging after every sub-cycle.

A third option came out of the measurements and had to be included: simply exchanging every K -th sub-cycle and letting the halo go stale. It is not exact, but it is by far the cheapest, so its accuracy had to be established rather than assumed.

Everything below is measured on production configurations. Where something is not measured, that is said.

Two points shape how the results below should be read. Every timing is given both as a share of the model step and as a share of the ice cost; the first divides an ice-only change by about ten, and reporting only it led us at first to describe your reformulation as having no effect, which it does not (Sect. 6.2). And the cheapest of the three schemes places the domain decomposition in the ice field (Sect. 6.4), so although it is the fastest we measured we do not regard it as usable.

2. Definitions, configurations and machine

The sub-cycle and the halo. mEVP advances the ice momentum with 120 sub-cycles per model time step in all our configurations. The mesh is divided among MPI processes; each process owns a piece of it and keeps a one-cell-wide copy (the *halo*) of its neighbours' edge values. In the standard scheme the ice-velocity halo is exchanged after *every* sub-cycle, so one model step costs 120 halo exchanges.

The five configurations. These names are used throughout.

name	carried between sub-cycles	halo exchange
standard	stress $\sigma_{11}, \sigma_{12}, \sigma_{22}$ at elements	after every sub-cycle
divergence form	$\nabla \cdot \sigma$ (two components) at nodes	after every sub-cycle
wide halo, standard form	stress at elements, ring K deep	every K -th sub-cycle, <i>exact</i>
wide halo, divergence form	$\nabla \cdot \sigma$ at nodes, ring K deep	every K -th sub-cycle, <i>exact</i>
delayed exchange	stress at elements	every K -th sub-cycle, <i>stale halo</i>

The two wide-halo configurations are exact in the strict sense: they reproduce the every-sub-cycle trajectory bit for bit. The delayed exchange does not, and section 6.5 says by how much.

Meshes and machine. All four are production configurations. Levante at DKRZ: GPU nodes carry four NVIDIA A100 (80 GB) and one MPI process per GPU; CPU nodes carry two AMD EPYC 7763, 128 cores each. Every comparison is made inside one job allocation with the same executable, and is the faster of two repetitions; repetition spread is below 0.4% on GPU and about 1.5% on CPU.

	surface nodes	elements	levels	time step
CORE2	126 858	251 746	48	1800 s
fArc	638 387	1 270 173	48	900 s
DARS	3 160 340	6 294 801	57	120 s
NG5	7 402 886	14 827 217	70	180 s

3. The cost of the sub-cycle

One model step of the standard scheme performs 120 ice-velocity halo exchanges, each moving very little data, with very little arithmetic between them. Koldunov et al. (2019, GMD 12, 3991) diagnosed this on the fArc mesh and proposed — without testing it — “wider halo regions to reduce communication frequency”. The wide halo above is that proposal.

We can now put numbers on the diagnosis in our own code.

On CPU, at 2304 cores on fArc, the ice part is 13.2% of the model step, and inside the ice dynamics the time spent *waiting* for MPI is 4.8 ms per step against 3.5 ms of actual computation. The cost is idle time, not arithmetic.

On GPU the same structure costs more, for a second reason. Each exchange is a sequence of small GPU operations — gather the values to be sent, copy, scatter the values received — and each carries a fixed issue cost independent of how little data it touches. On CORE2 with 8 GPUs the ice-dynamics phase takes 8.2 ms per model step, of which only about 1.25 ms is arithmetic, spread over roughly 480 separate GPU operations.

The mEVP sub-cycle as it stands is therefore limited by communication and by the number of separate operations dispatched, rather than by arithmetic or by memory traffic. That is the single fact behind most of what follows, including the results that came out negative.

4. State of the code before this work

FESOM2 (ocean and ice) has been rewritten in C++ with Kokkos and runs entirely on the GPU, with no per-step transfer of prognostic fields to the host. On CPU it is bit-for-bit identical to our validated C port of FESOM2, which was itself validated against Fortran FESOM2; on GPU it differs only at the level of summation order, checked against the CPU version over multi-decadal integrations. mEVP was already ported and validated, exchanging after every sub-cycle and keeping the stress at elements — that is, it was the “standard” row above, and it is the reference for everything that follows.

5. What was implemented

All four new configurations are selected at run time and are off by default, so the validated path is untouched when they are not requested.

The divergence form. The stress recursion is replaced by a recursion for $R = \nabla \cdot \sigma$ at nodes, with the ice-element mask applied so that R is advanced only where ice elements contribute.

The wide halo, both forms. At the start of each model step one extended exchange fills a ring of ghost nodes and ghost elements K cells deep with the owner’s values of the eleven fields the sub-cycle needs. Over the next $K - 1$ sub-cycles each process advances that ring itself, in the owner’s order, and needs no communication; on the K -th sub-cycle the ring is refreshed. The ring’s geometry, boundary mask and per-step element quantities are shipped from their owners rather than recomputed — a locally recomputed value can differ in the last bit, and that is enough to

destroy exactness.

The delayed exchange. One conditional: the exchange fires only when the sub-cycle index is a multiple of K . Nothing else changes.

How correctness was established. For the exact options the requirement is strict: with the option on, the model must produce *bit-for-bit* the same output as with it off. That was verified for the wide halo in both forms at $K = 4$ and $K = 8$, and separately for the same machinery under standard EVP. The options were also shown to leave the model unchanged when a different vertical mixing scheme or vertical coordinate is selected, and the GPU version was checked against the CPU version with the options on. The delayed exchange is an approximation, so bit-for-bit equality is not available; what it does to the ice is Sect. 6.4.

6. Results

The table below is the summary; the subsections give the measurements behind it. Each timing is quoted twice, against the model time step and against the ice cost (ice, ice dynamics and ice advection, computation together with time spent waiting for communication), for the reason set out in Sect. 6.2: sea ice is 5 to 13% of a CPU step, so an ice-only change is divided by about ten before it reaches the step.

		GPU		CPU	
		ice cost	model step	ice cost	model step
delayed exchange	$K=2$	−32 to −39%	−8.3 to −9.3%	−2 to −18%	−0.1 to −2.9%
delayed exchange	$K=4$	−37 to −58%	−8.5 to −14%	−5 to −25%	−0.3 to −3.6%
wide halo, standard form	$K=8$	−11 to −24%	−2.2 to −5.7%	never profitable	
wide halo, divergence form	$K=8$	+5 to −29%	+0.2 to −6.8%	never profitable	
wide halo, either form	$K=2$	−15% [†]	+1.7 to −3.8%	never profitable	
divergence form alone	—	+2 to +17%	0 to +1.6%	−2 to −8.5%	−0.5 to 0%

GPU ranges span the meshes at the node count where each still gains from more GPUs: CORE2 on 1 node (4 GPUs), fArc on 4, DARS on 8, NG5 on 16 — the wide-halo rows cover the first three, NG5 being still in the queue. CPU ranges span 128 to 2048 cores on CORE2 and fArc. [†] one point only (DARS, 8 nodes); at $K=2$ the wide halo is a cost on fArc at 4 nodes and a saving at 16, so it needs the larger K . The delayed exchange, which the table shows to be much the fastest, is ruled out on other grounds in Sect. 6.4.

6.1 Exactness of the reformulation

Carrying R and σ simultaneously and comparing them sub-cycle by sub-cycle, R agrees with $\nabla \cdot \sigma$ to $1\text{--}2.5 \times 10^{-15}$ relative, and this does not grow over 20 model steps, i.e. 2400 sub-cycles. The algebra survives the discretisation to round-off, not merely to plotting accuracy.

One genuine non-equivalence is worth naming. Within a model step the ice-element mask is fixed and the two forms are equivalent. *Across* steps, when an element enters or leaves the ice mask, they are not exactly equivalent, because in the divergence form the recursion has already been applied after summation over the elements around a node rather than before. We measured it: it fires only on steps where mask membership actually changes, and is bounded by 2.1×10^{-11} relative. That is far below physical significance, but it means the divergence form is not bitwise reproducible against the standard form over a long integration, and should be introduced as a scheme change rather than as an optimisation.

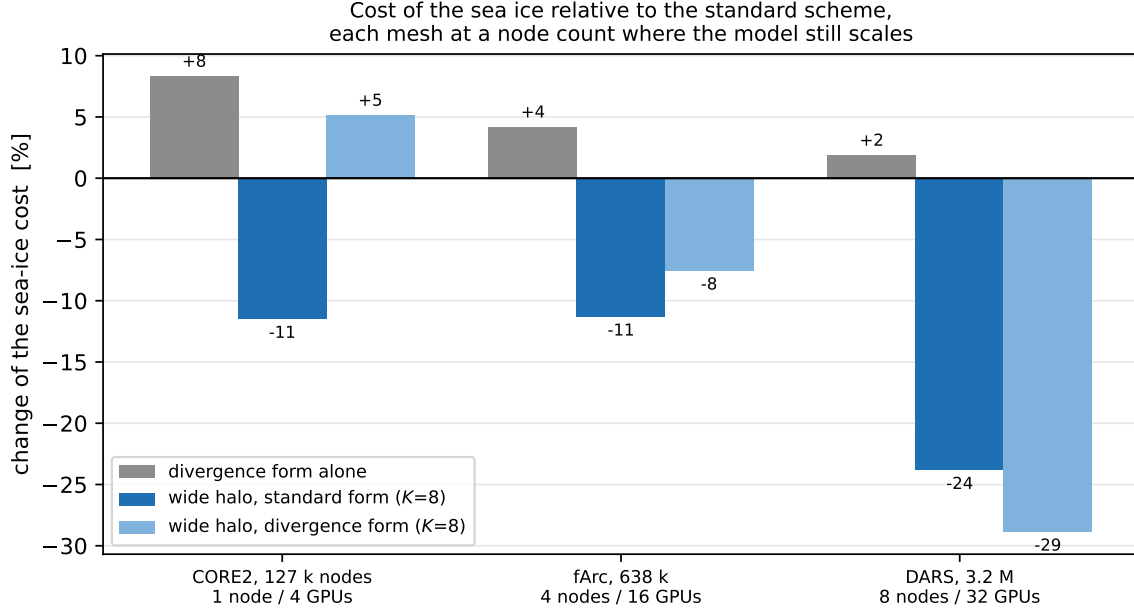


Figure 1: Cost of the sea ice relative to the standard scheme, for the schemes that survive Sect. 6.4. Negative is a saving. The measure is the ice cost — ice, ice dynamics and ice advection, computation together with time spent waiting for communication — and not the model time step, which divides an ice-only change by about ten. Each mesh is at its own operating point, the largest node count at which the model still gains from more GPUs, read off the measured strong-scaling curve of the standard scheme; each was measured in a single allocation with all configurations run back to back. Both wide-halo bars are exact and reproduce the standard trajectory bit for bit.

6.2 Effect of the reformulation on run time

Measured against the ice cost, the reformulation **does** pay on CPU, and the more so the more processes are used: -2.2% at 512 processes on fArc, -3.2% at 1024, -5.6% at 1536, -6.7% at 2048, and -8.5% on CORE2 at 512. Against the model step the same runs read -0.5% to 0% — which is why we first reported it as nothing.

On GPU it is a *cost* in both currencies: $+2$ to $+17\%$ of the ice cost, 0 to $+1.6\%$ of the model step (the ice-dynamics phase alone grows 13% on CORE2 with 8 GPUs, 7% on NG5).

The reason is section 3. To bound the largest saving the reformulation could possibly deliver we ran a control: three element-sized arrays that the mEVP path writes every sub-cycle and never reads were simply not written. That removes strictly more memory traffic than the reformulation does, and it changed the run time by **exactly 0.00%**. There was no memory-traffic saving available to collect. Profiling agrees from the other side: total GPU arithmetic time is identical in the two forms, 21.8 against 21.7 ms per step.

The verdict therefore differs between the backends rather than being uniformly null. The reformulation removes real work; on GPU that work was not being paid for, because there the sub-cycle is limited by how often work is dispatched rather than by arithmetic or memory traffic. On CPU, where that is not the case, it is worth having.

One caveat: the CPU numbers at the largest process counts are wait-dominated and noisy — at fArc with 2048 processes the ice cost falls 6.7% while the model step *rises* 1.0% , which is not internally consistent and is probably load-imbalance jitter. The CORE2 512-process point (-8.5% of the ice, -0.34% of the step, same sign) is the cleaner one. These deserve a re-measurement on matched instrumented pairs before they are quoted in a paper.

6.3 The wide halo

Figure 1 summarises what the wide halo removes from the ice cost. Every mesh is shown at *its own operating point* — the largest node count at which the model still gains from more GPUs, read off the measured strong-scaling curve. That matters: at a common large node count these numbers roughly double, but a configuration past its scaling limit is not one anybody runs, and quoting it would flatter every option in this study.

At those operating points the wide halo removes **11% of the ice cost on CORE2 and fArc, and 24 to 29% on DARS**, where the ice is a larger share of the step. The divergence form on its own adds 2 to 8% there, for the reason given in Sect. 6.2.

Reading the figure:

- The wide halo is a real gain, and it is exact, so no accuracy argument is needed: at the operating points it is worth -2.2 to -6.8% of the model step, which is **-11 to -29% of the ice cost** depending on the mesh and on which form carries it. For the comparison with your own figure: on fArc at 16 GPU nodes with $K=8$ the divergence form removes 19.1% of the ice cost and the classic form 11.9% — the same configuration and very nearly the same number as your $\sim 20\%$, the apparent disagreement having been a matter of denominator. On DARS at 8 nodes, an operating point rather than a flat one, it reaches 28.8%.
- It is worth substantially less than the delayed exchange on every mesh.
- On CPU it never wins, on any mesh, at any K .
- It needs large K . At $K = 2$ it is a net *cost* on most configurations, because the duplicated ring has not been amortised. This is the opposite of what the delayed exchange wants (Sect. 6.4), so the two options do not share a best K .

This reproduces, and sharpens, an earlier assessment. Our optimisation campaign measured the wide halo under standard EVP a month ago — NG5, 4 and 16 GPU nodes — and got $+1.7\%$ at $K = 4$ and -2.2% at $K = 8$, i.e. “low yield”. Re-measured this week at the same point with the current code: $+1.69\%$ and -2.26% . The earlier number was right. What that campaign also noted at the time, and never acted on, was that its wide refresh sent *two messages per neighbour* and that fusing them into one was a candidate fix; doing that now takes NG5 from -2.26% to -4.19% at $K = 8$ and from $+1.69\%$ to -2.47% at $K = 4$. So the fusion is worth roughly 2 to 4 percentage points — real, and not enough to change the verdict on NG5.

The CPU result has a clear mechanism. Trading communication for local recomputation is profitable only while the ring stays small relative to the sub-domain *and* a message stays expensive relative to the recomputation. The controlling quantity is the ratio of ghost cells that must be advanced to owned cells, which grows like $K \times (\text{perimeter/area})$ of the sub-domain. On GPU, where a process holds a large piece of the mesh, that ratio is 0.03 to 0.25 and the wide halo pays. On CPU, where processes are small and messages are comparatively cheap, it is 0.19 to 1.18 and it never did: on fArc with 2048 processes at $K = 4$ the ratio reaches 1.18 — the ring a process must advance is *larger than its own sub-domain* — and that configuration costs 5 to 7% instead of saving anything.

A second effect pushes the same way. The ring is a different size on every process, so duplicating it introduces a new load imbalance on top of the one it removes: on fArc CPU the ice-dynamics waiting time rose from 8.7 to 25.3 ms per step when the ring was made deep.

6.3b The same scheme, written as a widening of the existing loops

Everything above measures a wide halo whose ring computation runs in kernels of its own. That turns out to be most of what it costs.

The reference implementation we were sent extends the bounds of the loops that already exist and adds no kernels; ours launches a separate ring kernel per stage per sub-cycle. We profiled the difference before changing anything. Our wide halo adds **362 kernel launches per model step**, and each is on a slower launch path than the owned kernel it duplicates: the ring kernels capture enough array descriptors to cross the threshold in Kokkos above which arguments are staged through constant memory rather than passed with the launch. The measured gap between consecutive launches is $14.3\mu\text{s}$ for those against $4.4\mu\text{s}$ for the owned kernels, and over a step the ring kernels cost 5.9 ms of inter-kernel idle against 2.5 ms of arithmetic. The launch structure was more than twice the computation it carried.

Folding each ring kernel into the owned loop it duplicates — and, the two being one loop thereafter, assembling the ring nodes by the same scatter the owned nodes use instead of replaying the owner’s summation order — gives, on CORE2 at its operating point:

CORE2, one node, four GPUs, all six in one allocation	model step	sea-ice cost	
every sub-cycle (reference)	0.0667 s	15.7 ms	—
<i>delayed exchange (stale halo), $K=8$</i>	−14.4%	6.6 ms	−58.0%
wide halo, standard form, separate kernels	−3.0%	13.9 ms	−11.5%
wide halo, divergence form, separate kernels	+1.1%	16.1 ms	+2.6%
wide halo, standard form, widened loops	−14.1%	6.8 ms	− 56.7%
wide halo, divergence form, widened loops	−14.4%	6.5 ms	− 58.6%

Three points, in order of how much they change the study.

The trade between speed and a clean ice field does not exist. The delayed exchange is in the table because Sect. 6.4 rejects it for putting the decomposition into the ice field while recording that it saves about three times what the wide halo saves. It no longer does: in the divergence form the widened-loop wide halo matches it exactly on the model step and is 1.5% ahead of it on the ice cost, and in the standard form it is within a third of a percent. The wide halo still computes the ring and still refreshes it exactly every window, so it still leaves no imprint. The factor of three was our kernel structure, and with it goes the only argument this study had for a stale halo.

The two forms become indistinguishable. They differ by 3.5 percentage points with separate ring kernels and by nothing to four decimals with widened loops, although the divergence form still ships 40% fewer numbers. Sect. 6.5’s decomposition of that gap into a message term and a byte term is not withdrawn — both terms were isolated at bit-identical output — but what it decomposed was smaller than the implementation term sitting on top of it.

It is a compute result, not a communication one. The two rows of each pair post identical message and byte counts: 390.5 messages and 1 542 053 doubles per step for the standard form in both cases. Nothing moved on the wire.

The price is bit-identity, deliberately: a ring node is now summed in the receiving process’s element order rather than the owner’s. After a step of 120 sub-cycles the ice velocity differs from the exact wide halo by 8 to 15 units in the last place, and after twenty steps by about as much as this port’s own GPU and CPU arms differ at identical arithmetic. Since that removes the bit-identity gate, the 60-day boundary/interior test of Sect. 6.4 becomes the acceptance criterion for this variant rather than a diagnostic.

On CPU the same change is not a gain, and the reason is instructive: with no launches to remove it only trades a gather of plain reads for an atomic scatter. In the standard form, where the gather merely re-read stored stresses, that costs 5 percentage points at 512 cores on fArc. In the divergence form, where the gather had to recompute each element’s stress once per incident ring node, about six times over, it saves 3.

6.4 The delayed exchange, and why it is not usable

Accepting a halo up to $K - 1$ sub-cycles old removes 105 of the 120 exchanges per step at $K = 8$ and costs one conditional. At the four operating points it takes -14.4 , -14.3 , -15.5 and -9.9% off the model step at $K = 8$ (CORE2, fArc, DARS, NG5) and -12.6 , -11.4 , -14.2 and -8.5% at $K = 4$, which against the ice cost is -37 to -58% on GPU and -5 to -25% on CPU. On CPU the same change is worth -1 to -3.4% of the step, because a CPU process holds far less of the mesh. *We described this as about three times what the wide halo gives; Sect. 6.3b withdraws that comparison. Measured against the wide halo written as a widening of the existing loops, in the same allocation, the delayed exchange is level on CORE2 and marginally behind on the ice cost.*

We do not regard it as usable. The stale values lie on the one-cell boundary of each sub-domain, and the decomposition is fixed for the run, so the same nodes are perturbed at every step. Figure 2 shows the consequence after 60 days on fArc: the difference follows the sub-domain boundaries, while the exact wide halo on the same colour scale does not. Averaged on those boundaries the difference in ice thickness is 2.5 times its interior value at $K = 4$ and 1.7 times at $K = 2$, against 0.85 for two identical runs and 0.87 for the exact wide halo. A polygonal structure in the ice field, in the shape of the domain decomposition, is not acceptable in a sea-ice model, and that ends the matter.

Worth recording is that no integral diagnostic sees it. Over a year on CORE2 the ice concentration and thickness correlate with the reference at 0.999995 and 0.999994, within the spread between our C port and Fortran FESOM2, and hemispheric ice extent departs by 0.23% of its seasonal amplitude. A feature one cell wide along sub-domain edges covers a small fraction of the mesh and leaves a domain-wide correlation near 0.99999. The accuracy protocol we had prepared, and had already run, would have passed this scheme.

The measurement also separates the two ways of giving up bit-identity that your note treats together. Recomputing a shared quantity independently, so that each core may change the last bit on a triangle it shares, is harmless: two runs differing only in that respect are our control, at 0.85. Reading a halo that is $K - 1$ sub-cycles old introduces an error of physical rather than rounding magnitude, at 2.52. The wide halo recomputes; the delayed exchange lets values age. Only the second imprints the decomposition.

6.5 The message count, and a correction to an earlier attribution

This is the result we would most like your view on, because it corrects something we believed until two days ago.

The two wide-halo forms differ in what they must send. The standard form must send node velocities *and* the ghost ring's element stresses — in our implementation, two messages per neighbour. The divergence form sends one message, because everything it carries lives at nodes, and it sends about half as many numbers. We measured the divergence form to be the faster of the two on every GPU mesh, by up to 7 percentage points of the model step, and attributed that to the smaller volume — which is the advantage the reformulation is designed to deliver.

That attribution was wrong. We changed the standard form to pack its two pieces into a *single* message per neighbour — identical values in identical order, verified bit for bit against the two-message version — and the difference reversed. Holding the volume fixed and removing one message is worth up to 8.6 percentage points of the model step; holding the message count fixed and removing 19% of the data is worth nothing. On DARS with 32 GPUs the two configurations then send *exactly* the same number of messages (24 400 per step) while the divergence form sends 19% fewer numbers — and the divergence form is 1.3 percentage points *slower*, because it must reconstruct the ghost ring's stresses in an extra pass.

Across ten configurations where all three variants ran side by side, removing the message helps

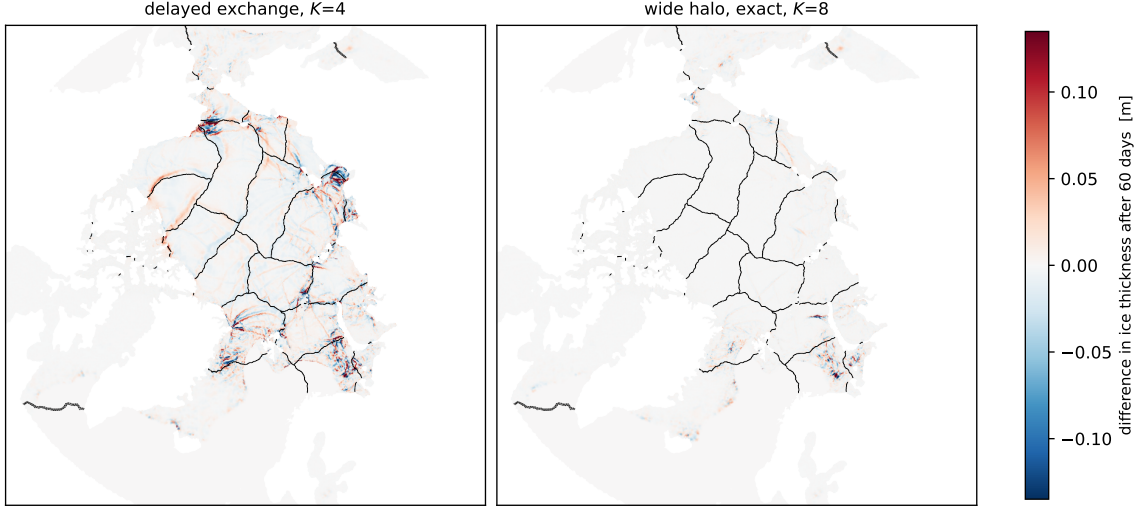


Figure 2: Difference in ice thickness against a run exchanging after every sub-cycle, fArc on 16 GPUs after 60 days, with the 16 sub-domain boundaries drawn on top and a common colour scale. The delayed exchange (left) leaves a difference that follows the boundaries; the exact wide halo (right) does not, and differs from the reference only in the order of summation.

everywhere (-0.1 to -8.6 percentage points), while halving the data is worth between -1.3 and $+2.8$ points and favours the *standard* form in eight of the ten.

The conclusion is not that the reformulation is wrong — section 6.1 shows it is exact — but that on this hardware the cost of an exchange is dominated by the per-message overhead rather than by its volume, so halving the volume buys nothing that the extra local pass does not immediately give back. Across fourteen configurations the two forms now sit within about 1.5 percentage points of each other, with the standard form ahead in ten of them; where the divergence form wins — DARS and NG5 — it wins by about one point. Neither is systematically better.

7. Summary of the three schemes

	status	our reading
divergence form	exact to round-off; on CPU -2 to -8.5% of the ice cost, growing with process count; on GPU a cost of $+2$ to $+17\%$	Not without effect: our first report of it as such came from quoting the model step, in which the ice is a tenth of the total. It removes real work, but on GPU that work was not being paid for.
wide halo (either form)	exact, bit for bit; -12 to -19% of the ice cost on GPU at $K=8$ (-2 to -7% of the model step); never profitable on CPU; no partition imprint	The one we would put into production: the only scheme here that is both worth having and free of both an accuracy argument and an artefact. Its -19.1% of the ice cost on fArc is the same measurement as your $\sim 20\%$.
delayed exchange	approximate; -37 to -58% of the ice cost on GPU — three times the wide halo — but it draws the sub-domain boundaries into the ice (edge/interior 2.52 at $K=4$ against a control of 0.85, and still 1.70 at $K=2$)	The fastest scheme in this study, and one we do not consider usable: a polygonal artefact in the shape of the domain decomposition disqualifies a sea-ice scheme however small the integral error is. Reported here as a negative result.

8. Not measured

- **The accuracy of the delayed exchange at high resolution.** This was to have been the largest remaining gap; Sect. 6.4 rules the scheme out on other grounds, so the one-year CORE2 result stands as a record of what the approximation costs rather than as support for using it.
- **Trimming the eleven auxiliary fields.** Measured but not built. Instrumenting the per-step exchange to compare each arriving owner value against the value the receiving process already holds shows that *nine of the eleven arrive exactly equal* — ice velocity, concentration, thickness, snow, ocean velocity, wind stress — because FESOM’s own halo exchange has already put them there. The two right-hand sides do not, being assembled and area-scaled over owned nodes only. But the redundancy stops at the first ring, and rings $2 \dots K$ have no other source at any K , so what could be removed is 37% of that exchange at $K=2$ and 17% at $K=4$; since the exchange is 5.4% of the data the scheme moves and one message per neighbour per step, removing it would gain under one percent of the traffic and no messages at all. We did not implement it, and this is the measurement that says why.
- **The widened-loop variant on the three larger meshes.** Sect. 6.3b is CORE2 at its operating point; fArc, DARS and NG5 are queued at the time of writing. The claim that the implementation rather than the scheme carried the cost rests on one mesh until they land.
- **The divergence form over a long integration.** Since it showed no speed benefit we did not carry it through a one-year test, so the 2.1×10^{-11} per-step non-equivalence of section 6.1 has not been followed to its long-term consequences.
- **Core counts above 2304 on fArc.** The 2019 paper reaches 6912. The partition set we have for that mesh stops at 2304, and the separately partitioned copy in the shared archive does not run in this port (the ocean solver fails to converge immediately, at every time step tried, which is a configuration problem rather than a stability one). Our measured ice share reaches 13% at 2304 cores where the paper interpolates 25–30%; the two candidate reasons — this port’s own communication optimisations, and mEVP here versus standard EVP there — are untested.
- **The 13% GPU cost of the divergence form** (section 6.2) is measured but not explained. It is not arithmetic time and not the masked node kernel; we have no mechanism for it and are not offering one.

All configurations are switchable at run time in the port; the underlying runs, job scripts and the full internal write-up can be shared on request.